

Dimostrare che non esiste il limite $\lim_{x \rightarrow +\infty} \underbrace{(x + x^3 \sin x)}_{f(x)}$

$$a_n = 2n\pi \longrightarrow +\infty, \quad \sin(a_n) = 0$$

$$b_n = -\frac{\pi}{2} + 2n\pi \longrightarrow +\infty, \quad \sin b_n = -1$$

$$f(a_n) = a_n = 2n\pi \xrightarrow{n \rightarrow +\infty} +\infty$$

$$f(b_n) = b_n - b_n^3 = b_n \underbrace{(1 - b_n^2)}_{- \infty} = -\infty$$

$\downarrow$
 $+\infty$

$\downarrow$
 $-\infty$

$$\lim_{n \rightarrow +\infty} f(a_n) \neq \lim_{n \rightarrow +\infty} f(b_n)$$

$$\Rightarrow \lim_{x \rightarrow +\infty} f(x) \nexists \text{ per il 1. funz. composta}$$